

DISCORD GRAND PRIX

UNIFIED MASTER ENGINE SPECIFICATION & FEATURE GUIDE

All Bot Features | Deep Mathematical Models | Physics Equations
Admin Suite | 24 Official F1 Track Profiles | 45 Commands

SYSTEM CORE SUB-SYSTEMS:

- 6-Part Garage & Upgrade System (Levels 1-20)
- Crate Unboxing, Parts Inventory & 5-Tier Rarity Offsets
- Driver Skills Training & Dynamic Track Mastery familiarization
- 1v1 Duels, Wager Challenges & full 20-Car Grand Prix weekends
- Qualifying Knockouts (Q1 -> Q2 -> Q3) & Live Telemetry
- Advanced Physics Engine (tyre wear, non-linear degradation, mechanical DNFs)
- World Driver Championship (WDC) Season Manager & automated calendar
- Modular Python Cogs architecture built with SQLite WAL-mode databases

DOCUMENT SPECIFICATION

Version: 2.0 | Updated: July 2026

Database Schema: Active Build V2 | Target: discord.py 2.x API

[CAR] DISCORD GRAND PRIX

UNIFIED MASTER ENGINE SPECIFICATION & FEATURE GUIDE

All Bot Features - Deep Mathematical Models - Physics Equations - Admin Suite - Track Profiles

DISCORD GRAND PRIX is an advanced, high-performance Formula 1 esports and team management engine built for Discord. Combining deep RPG progression, realistic telemetry physics equations, a full World Driver Championship (WDC) calendar management system, an item economy with crate unboxing, and automated server activity rewards -- it delivers the ultimate racing simulation experience.

[NOTE] This guide reflects the **current live build** of the bot as of July 2026. Features marked with [SPRINT] are new since the original specification.

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1. ALL GAMEPLAY & RPG FEATURES

Feature Subsystem	Capabilities & Functionality Overview
RPG Team & Driver Training	Train 6 Driver Skills (Pace, Qual, Wet, Consistency, Aggression, Overtaking) with /train. Each skill ranges from 1 - 100. Training costs 400¢ per session.
6-Part Garage & Damage Engine	Upgrade & manage Engine, Aerodynamics, Tyres, ERS, Reliability, & Pit Crew (Levels 1 - 20). Dynamic damage accumulates per race with realistic power loss penalties. Repair with /repair.
[SPRINT] Crate & Inventory System	Unbox Rookie (500¢), Pro (2,500¢), or Champion (6,000¢) crates. Drop parts across 5 rarity tiers (Common -> Legendary) with base level offsets and efficiency multipliers. Equip parts from /inventory.
[SPRINT] Consumable Boosters Shop	Purchase tactical boosters from /shop -- Tyre Blanket Warmer, ERS Injector, or Heavy Duty Radiator. Max 2 active boosters at a time.

Feature Subsystem	Capabilities & Functionality Overview
Economy & Passive Rewards	Earn credits through /daily (500¢), /work (250 - 600¢), chat activity (25¢/msg, 1,000¢/day cap), and voice channel time (15¢/min, 1,500¢/day cap).
1v1 Fast-Paced Duels	Challenge rivals to head-to-head duel races (/race) with live lap-by-lap telemetry updates, dynamic overtaking, and real-time strategy adjustments.
[SPRINT] High-Stakes Wager Duels	Challenge rivals with credit wagers using /bet. Winner takes the full doubled pot. Both players must have sufficient balance.
Grand Prix Events (20-Car Grid)	Admin-scheduled full GP weekends with Q1/Q2/Q3 qualifying, dynamic weather, Safety Car/VSC, DRS overtaking, and F1-standard point distribution (25-18-15-12-10-8-6-4-2-1).
[SPRINT] Sprint Race Weekends	Shorter sprint format races with alternate points distribution (8-7-6-5-4-3-2-1 for Top 8). Official sprint tracks marked in the calendar.
[SPRINT] Track Mastery & Practice	Run solo practice sessions (/practice) on any official F1 track. Earn pace bonus up to -0.15s/lap. Max 3 sessions per day. Costs 500¢ per session.
WDC Season Championship	Full World Driver Championship management. Create multi-round seasons, schedule GP & Sprint rounds, view live standings, and crown a World Champion.
Leaderboard System	View server-wide rankings sorted by Championship Points, Total Wins, Team Wealth, or Driver Level via /leaderboard.

2. ONBOARDING & TEAM CREATION

/start -- Initialize Your Racing Team

Every player begins their journey by creating a unique team profile.

Starting Resources: | Resource | Starting Value | | Credits (Money) | 1,500¢ | | Driver Level | 1 | | Experience (XP) | 0 | | All Driver Skills | 50 / 100 | | All Garage Parts | Level 1 / 20 | | Damage | 0% |

Team Name Rules: - Must be 3 - 32 characters long - Must contain at least one letter - Allowed characters: letters, numbers, spaces, hyphens, underscores, apostrophes - Must be unique per server (case-insensitive)

Post-Creation: - A driver record is initialized (all skills at 50) - A strategist record is initialized (all skills at 50) - A garage record is initialized (all parts at Level 1)

/profile -- View Your Team Profile Card

Renders a premium **880x540px F1-style PNG profile card** featuring: - Carbon-teal header gradient with team name and country flag - Driver Level, Overall Power rating, and Equipped Rarity Class - Credits, XP progress bar, Win/Loss record - Left column: 6 Driver skill bars (1 - 100 scale, color-coded) - Right column: 6 Garage component bars with equipped part rarity tags

/team -- Detailed Team Summary

Shows an embed summary of a team including all driver, strategist, and garage stats.

3. 6-PART GARAGE & UPGRADE SYSTEM

The garage contains **6 upgradeable car components**, each on a **Level 1 - 20** scale. Higher levels provide proportionally more race performance.

Component Overview

Component	Weight in Car Power	Race Impact
[WRENCH] Engine	1.0x (Highest)	Raw straight-line speed. Shaves 0.05s/lap per level.
[WRENCH] Aerodynamics	0.8x	Corner speed & downforce. Shaves 0.04s/lap per level.
[WRENCH] Tyres	0.5x	Grip & tyre longevity. Reduces tyre wear impact.
[WRENCH] ERS System	0.8x	Energy recovery & deployment. Shaves 0.04s/lap per level.
[WRENCH] Reliability	0.25x	Reduces DNF/mechanical failure probability.
[WRENCH] Pit Crew	0.25x	Reduces pit stop stationary time.

Upgrade Cost Table (Levels 1->20)

Costs scale exponentially. Each part type uses a **cost multiplier** relative to the Engine base cost:

Level	Engine (1.0x)	Aero / ERS (0.8x)	Tyres (0.5x)	Reliability / Pit Crew (0.25x)
2	200¢	160¢	100¢	50¢
3	400¢	320¢	200¢	100¢
4	700¢	560¢	350¢	175¢
5	1,100¢	880¢	550¢	275¢
6	1,600¢	1,280¢	800¢	400¢
7	2,200¢	1,760¢	1,100¢	550¢
8	2,900¢	2,320¢	1,450¢	725¢
9	3,700¢	2,960¢	1,850¢	925¢
10	4,700¢	3,760¢	2,350¢	1,175¢
11	5,900¢	4,720¢	2,950¢	1,475¢
12	7,300¢	5,840¢	3,650¢	1,825¢
13	8,900¢	7,120¢	4,450¢	2,225¢
14	10,700¢	8,560¢	5,350¢	2,675¢
15	12,700¢	10,160¢	6,350¢	3,175¢
16	15,000¢	12,000¢	7,500¢	3,750¢
17	17,500¢	14,000¢	8,750¢	4,375¢
18	20,300¢	16,240¢	10,150¢	5,075¢
19	23,400¢	18,720¢	11,700¢	5,850¢
20	26,800¢	21,440¢	13,400¢	6,700¢

[SPRINT] Level Tier Stat Scaling System

Higher-level components gain exponentially better efficiency through a **4-Tier multiplier system**:

Tier	Level Range	Stat Efficiency Multiplier
Tier 1 (Spec/Rookie)	Levels 1 - 5	1.00x (100%)
Tier 2 (Performance)	Levels 6 - 10	1.25x (125%)
Tier 3 (Advanced)	Levels 11 - 15	1.60x (160%)
Tier 4 (Elite/Extreme)	Levels 16 - 20	2.00x (200%)

Damage & Repair System

Cars accumulate damage after every race: - **Finishing**: Engine +3 - 10%, Tyres +10 - 25% - **DNF (Crash)**: Engine +15 - 30%, Tyres +30 - 60% - **Damage Penalty**: -1% car power per 10% total damage (capped at -50%)

Repair Costs: 25¢ per 1% damage

Command	Description
/repairs	View current damage levels and repair cost estimates
/repair engine	Repair engine damage to 0%
/repair tyres	Repair tyre damage to 0%
/repair full	Full car overhaul -- repairs all damage

4. DRIVER PERSONNEL TRAINING

Train 6 driver skills from Level 50 -> 100. Each training session costs **400¢** and increases one skill by +1.

Skill	Race Impact
Pace	Raw lap time speed. Shaves up to 2.0s/lap at maximum (100).
Qualifying	One-lap qualifying speed bonus. Higher = better grid positions.
Wet Skill	Performance in rain. Reduces wet weather lap time penalty significantly.
Consistency	Reduces random variance in lap times and lowers mistake/lockup probability.
Aggression	Increases overtake attempt frequency and provides minor pace boost, but increases crash risk and tyre wear.
Overtaking	Improves DRS pass success rate. Directly competes against defender's Consistency stat.

Command: /train <skill>

5. STRATEGIST PERSONNEL

Each team has a Strategist with 6 skills stored in the database:

Skill	Purpose
Pit Timing	Quality of pit window strategy calls
Weather Call	Accuracy of weather prediction and tyre change timing
Undercut	Ability to gain positions through pit stop timing
SC Skill	Safety Car restart performance
Risk	Risk management assessment
Communication	Engineer-to-driver radio clarity

[NOTE] Strategist skills are stored in the database but currently influence race outcomes through the overall team strategy system rather than individual stat checks.

6. CRATE UNBOXING & INVENTORY SYSTEM

Crate Tiers

Crate	Price	Gold Refund Range	Part Drop Chance	Available Rarities
[BOX] Rookie Crate	500¢	25 - 125¢	60%	Common (70%), Uncommon (25%), Rare (5%)
[CASE] Pro Crate	2,500¢	125 - 625¢	85%	Uncommon (40%), Rare (45%), Epic (12%), Legendary (3%)
[TROPHY] Champion Crate	6,000¢	300 - 1,500¢	100% [OK]	Rare (35%), Epic (50%), Legendary (15%)

Commands: /crate (view), /open <tier> (unbox)

Part Categories Droppable

Parts drop for one of 6 car categories: **Engine, Aerodynamics, Tyres, ERS, Reliability, Pit Crew**

Each part has a unique name drawn from a curated pool per category (e.g., "V6 Turbo Power Unit", "Active Aero Rear Wing", "Ultra-Capacitor Bank").

[SPRINT] Rarity Base Level Offsets

Higher rarity parts carry **intrinsic base power boosts** -- a Level 1 Rare part performs like a Level 7 Common part:

Rarity	Emoji	Base Level Offset	Efficiency Multiplier
(Common) Common	(Common)	+0 levels	1.00x
(Uncommon) Uncommon	(Uncommon)	+3 levels	1.05x (+5%)
(Rare) Rare	(Rare)	+6 levels	1.12x (+12%)
(Epic) Epic	(Epic)	+10 levels	1.22x (+22%)
(Legendary) Legendary	(Legendary)	+15 levels	1.35x (+35%)

[SPRINT] Effective Stat Calculation

```
Effective_Level = Base_Level + Rarity_Offset
Tier_Multiplier = get_tier_stat_multiplier(Effective_Level)
Bonus_Multiplier = Rarity_Bonus_Multiplier
```

```
Final_Stat = Effective_Level x Tier_Multiplier x Bonus_Multiplier
```

Example: A Level 5 Legendary Engine: - Effective Level = 5 + 15 = **20** - Tier Multiplier = **2.0x** (Tier 4) - Bonus Multiplier = **1.35x** - Final Stat = 20 x 2.0 x 1.35 = **54.0** (vs. Level 5 Common = 5 x 1.0 x 1.0 = 5.0 -- a **10.8x advantage**)

Inventory Management

Command	Description
/inventory	View all stored parts, equipped items, and rarity details
Select an item -> Equip	Equip a part onto your active car setup (replaces current equipped item in that category)

7. CONSUMABLE BOOSTERS SHOP

Purchase single-use tactical boosters for qualifying & race sessions.

[LOCK] **Inventory Cap:** Maximum **2 active boosters** at any time.

Booster	Price	Effect
[FIRE] Tyre Blanket Warmer	1,500¢	-0.15s Qualifying Lap Pace advantage
[SPRINT] ERS High-Flow Injector	2,000¢	+1 Lap extra ERS boost during races
[SHIELD] Heavy Duty Radiator	1,200¢	-30% engine thermal heat buildup

Commands: /shop (view), /booster <item> (purchase)

8. ECONOMY & EARNING CREDITS

Active Earning Methods

Method	Command	Reward	Cooldown
Daily Login Bonus	/daily	500¢ flat	Once per day
Team Work Job	/work	250 - 600¢ random	Once per day
1v1 Duel Win	/race	200¢ (winner) / 50¢ (loser)	60s cooldown
Wager Duel Win	/bet	Winner takes 2x wager	--
GP Podium Finish	(Auto)	P1: 5,000¢ - P2: 3,000¢ - P3: 1,500¢	Per GP
GP Participation	(Auto)	500¢ base reward	Per GP
AI Rival Defeat	(Auto)	+500¢ bonus if you beat your AI rival	Per GP

Passive Earning Methods

Activity	Rate	Daily Cap
Chat Messages	25¢ per message	1,000¢/day (40 messages)
Voice Channel	15¢ per minute	1,500¢/day (100 minutes)

[IMPORTANT] Voice credits require being **actively unmuted and undeafened** in a voice channel. Self-muted, server-muted, or deafened members do not earn credits. A background task checks and awards voice credits every 5 minutes.

XP & Level Progression

- **Level Up Threshold:** Level $\times$ 1,000 XP
- **Level Up Reward:** New Level $\times$ 500¢ bonus credits
- **XP Sources:**
 - 1v1 Duel: Win = 100 XP, Loss = 25 XP
 - GP Completion: 500 XP (1,500 XP for podium finish)

[SPRINT] GP Stat Boost System

Finishing in the top 8 of a Grand Prix grants **direct driver skill boosts** to all 6 skills:

Position	Stat Boost (All 6 Skills)
P1	+8 to each
P2	+7 to each
P3	+6 to each
P4	+5 to each
P5	+4 to each
P6	+3 to each
P7	+2 to each
P8	+1 to each

9. RACING MODES

Mode 1: 1v1 Duel Race (/race)

- Challenge another player to a head-to-head duel
- Configurable lap count (default: 1 lap)
- Random or specified track selection
- Live lap-by-lap telemetry with interactive strategy buttons
- Mid-race strategy changes (Aggressive/Balanced/Conservative)
- Manual pit stop scheduling via telemetry view

Rewards: Winner: +200¢ & +100 XP - Loser: +50¢ & +25 XP

Mode 2: Wager Duel (/bet)

- Same as 1v1 duel but with a credit wager
- Both players must have enough credits to cover the wager
- Winner takes the entire doubled pot (2x wager amount)

Mode 3: Grand Prix Event (Admin-Scheduled)

- Full 20-car grid (human players + AI fill)
- 3-phase qualifying (Q1 -> Q2 -> Q3)
- Dynamic weather with radar forecasting
- Safety Car & Virtual Safety Car mechanics
- F1-standard points (25-18-15-12-10-8-6-4-2-1)
- Fastest Lap bonus point
- Entry fee: 1,000¢

Mode 4: Sprint Race Weekend (Admin-Scheduled)

- Shorter race format (default 8 laps vs. 15 for GP)
- Alternate points distribution (8-7-6-5-4-3-2-1)
- Official sprint tracks: Shanghai, Miami, Red Bull Ring, Austin, Interlagos, Qatar

10. GRAND PRIX RACE WEEKEND

Race Lifecycle

[FLOW DIAGRAM: WDC Season and Championship Lifecycle (Details in guide command registry)]

Registration Phase

- Players join with `/joinrace` (costs 1,000¢ entry fee)
- Players leave with `/leaverace` (full refund before race starts)
- View registered drivers with `/grid`

Weather System

Weather is generated per-race and stored as a **per-lap timeline**:

Weather Type	Generation Probability	Characteristics
[SUNNY] Sunny	50%	Clear conditions. 15% chance of a brief 2-lap drizzle.
[RAIN] Rain	20%	Dynamic patterns: starts wet -> dries, starts dry -> gets wet, or mid-race wet spell.
[SUN] Mixed	30%	Dry start with rain showers in the middle of the race.

Weather Radar: The bot broadcasts weather warnings: - **2 laps ahead:** Radar prediction of incoming weather change - **1 lap ahead:** Imminent weather change alert - **Current lap:** Weather change confirmation

Safety Car & Virtual Safety Car

Trigger	SC Chance	VSC Chance
Crash DNF	60%	40%
Mechanical DNF	--	25%
Random Debris	40% x track SC multiplier	60% x track SC multiplier

Safety Car Effects: - Lap time forced to $T_{base} \times 1.6$ (slow parade pace) - Tyre wear reduced to **10%** of normal - Gap compression: All cars bunched to **max 0.5s gaps** - Duration: 4 laps - Free pit stops under SC cost only 3.0s + crew time (vs. 8.0s + crew time normally)

Virtual Safety Car Effects: - Lap time forced to $T_{base} \times 1.3$ - Tyre wear reduced to **30%** of normal - Duration: 3 laps

Pit Stop Mechanics

Base Pit Stop Time:

$T_{pit} = 8.0s \text{ (pit lane loss)} + Crew_Time$
 $Crew_Time = 3.5s - (Pit_Crew_Level \times 0.15s)$

Under Safety Car:

$T_{pit} = 3.0s + Crew_Time$ (cheap pit stop!)

Pit Strategy Commands: | Command | Description | |---| | /strategy | Set starting pacing mode and tyre compound | | /pit <tyre> | Schedule pit stop for next lap of active GP |

Custom Pit Strategy JSON: Players can configure detailed pit stop plans through /strategy, including: - Starting tyre compound - Planned pit stop laps - Tyre choice per pit stop - Pacing mode per stint

DRS & Overtaking System

Overtaking is simulated when a trailing car is within **0.6s** of the car ahead:

$P(\text{Overtake}) = 0.45 + (\text{Overtaking} - \text{Consistency}) / 200$
 + Aggression / 250
 + Tyre_Advantage
 + Car_Power_Advantage

Where: - Tyre_Advantage = $(\text{back_tyre_health} - \text{front_tyre_health}) / 100$ (capped +/-0.25 to +0.40) -

Car_Power_Advantage = $(\text{back_car_power} - \text{front_car_power}) / 50$ (capped +/-0.20 to +0.30) - Final probability clamped to **10% - 95%**

Points Distribution

Position	GP Points	Sprint Points
P1	25	8
P2	18	7
P3	15	6
P4	12	5
P5	10	4
P6	8	3
P7	6	2
P8	4	1
P9	2	--
P10	1	--
Fastest Lap	+1 (if Top 10)	+1 (if Top 10)

11. QUALIFYING SYSTEM (Q1/Q2/Q3)

Full 3-phase knockout qualifying simulated authentically:

Qualifying Lap Time Calculation

$Lap_Time = Base_Track_Time - (Performance \times 0.12) + Random_Variance(+/-0.3s)$

$Performance = Car_Power + (Qual_Skill / 2) + Tyre_Bonus$

Tyre Compound Qualifying Bonus: | Compound | Bonus | |---| | Soft | +6.0 | | Medium | +3.0 | | Hard | +0.0 |

Command: /strategy allows setting qualifying tyre compound before each session.

12. RACE SIMULATION PHYSICS ENGINE

Base Lap Time Formula

```

T_lap = T_base
- (Engine x 0.05 + Aero x 0.04 + ERS x 0.04)    <- Car Bonus
- (Pace / 100 x 2.0)                             <- Driver Pace Bonus
- Aggression_Pace_Boost                          <- (Aggression - 50) / 10
+ Consistency_Variance                          <- Random +/- 0.4 x (1.5 - Consistency/100)
+ Strategy_Delta                                <- Aggressive: -0.8s / Conservative: +1.0s
+ Wear_Penalty                                  <- Non-linear tyre degradation
+ Weather_Penalty                               <- Wrong tyres in rain/dry

```

Tyre Wear Curve

$\text{Wear_per_lap} = \text{Random}(\text{Base} \pm 1.5) \times \text{Strategy_Mult} \times \text{Aggression_Wear_Mult}$

Base Wear Rates:
 Soft = 12%/lap
 Medium = 7%/lap
 Hard = 4%/lap

Strategy Multipliers:
 Aggressive = 1.4x
 Balanced = 1.0x
 Conservative = 0.7x

Aggression Wear: $1.0 + (\text{Aggression} - 50) / 250$

Wear Penalty (Non-Linear Degradation)

When tyre health drops below **80%**:

$\text{Wear_Penalty} = ((80 - \text{Tyre_Health})^{1.5}) \times 0.005 \text{ seconds}$

Tyre Health	Penalty
80%	0.000s
60%	0.448s
40%	2.530s
20%	6.573s
0%	12.649s

Weather Penalty

Condition	Penalty
Rain + Dry Slicks	15.0 - (Wet_Skill / 10) seconds
Sunny + Intermediates	5.0s + extra 15% tyre wear
Correct Tyres	0.0s

Damage Pace Penalty

$\text{Damage_Multiplier} = 1.0 - \min(0.5, \text{Total_Damage} / 1000)$
 $\text{Car_Power} = \text{Base_Car_Power} \times \text{Damage_Multiplier}$

DNF (Did Not Finish) Probability

$\text{DNF_Base_Chance} = \max(0.2, 6.5 - \text{Reliability} \times 0.25)$

$\text{Aggression Multiplier} = 1.0 + (\text{Aggression} - 50) / 100$

Strategy Modifier:

Aggressive = +3.0%

Conservative = -2.0% (minimum 0.05%)

Rain Multiplier:

Dry slicks on wet track = 3.5x

Correct wet tyres = 1.5x

$\text{Per_Lap_Chance} = \text{Final_Chance} / \text{Total_Laps}$

Driver Mistake (Spin/Lockup)

$\text{Spin_Probability} = 0.02 \times (1.5 - \text{Consistency} / 100)$ per lap

$\text{Spin_Time_Loss} = \text{Random}(8.0 - 15.0)$ seconds

13. TRACK MASTERY & PRACTICE SESSIONS

/practice -- Solo Practice Session

Run practice sessions on any official F1 track to build **Track Familiarity**.

Parameter	Value
Cost	500¢ per session
Daily Limit	3 sessions per day (across all tracks)
Pace Bonus per Session	+0.04s/lap improvement
Maximum Bonus	-0.15s/lap (100% Track Mastery)
Sessions to Max	4 sessions on a single track

14. OFFICIAL 24 F1 TRACK PROFILE MATRIX

Every track has unique modifiers that amplify or reduce the importance of specific car components.

Circuit	Description	Engine	Aero	Tyre	SC Mult	Sprint?
Bahrain (Sakhir)	Heavy braking & traction. Severe rear tyre wear.	1.25x	0.90x	1.40x	1.00x	[X]
Jeddah (Saudi Arabia)	Ultra-fast street circuit. Extreme barrier crash risk.	1.40x	1.20x	1.00x	1.70x	[X]
Melbourne (Albert Park)	High-speed flowing street layout with tricky braking zones. Legendary figure-8.	1.20x	1.10x	1.10x	1.00x	[X]
Suzuka (Japan)	Demands extreme aero & driver consistency.	1.10x	1.50x	1.00x	1.00x	[X]
Shanghai (China)	Monster 1.2km straight & technical hairpins.	1.35x	1.00x	1.25x	1.00x	[OK]
Miami (USA)	High heat street circuit around Hard Rock Stadium.	1.20x	1.00x	1.30x	1.00x	[OK]

Circuit	Description	Engine	Aero	Tyre	SC Mult	Sprint?
Imola (Emilia Romagna)	Historic narrow track. High downforce, difficult overtaking.	1.10x	1.40x	1.00x	1.00x	[X]
Monaco (Monte Carlo)	The Crown Jewel. Ultra-narrow, Aero & Quali dominate.	0.70x	1.70x	1.00x	1.00x	[X]
Montreal (Canada)	Stop-and-go circuit. Wall of Champions.	1.35x	1.00x	1.00x	1.60x	[X]
Barcelona (Catalunya)	Ultimate aero testbed. High tyre thermal degradation.	1.10x	1.50x	1.40x	1.00x	[X]
Red Bull Ring (Austria)	Short, power-heavy track in Styrian mountains.	1.45x	1.00x	1.00x	1.00x	[OK]
Silverstone (Great Britain)	High-speed sweeping curves. Extreme tyre load.	1.20x	1.50x	1.50x	1.00x	[X]
Hungaroring (Hungary)	Twisty 'Monaco without walls'. High downforce required.	1.00x	1.50x	1.00x	1.00x	[X]
Spa-Francorchamps (Belgium)	Eau Rouge power track. High rain chaos.	1.45x	1.25x	1.00x	1.00x	[X]
Zandvoort (Netherlands)	Banked corners & coastal dunes. High aero grip.	1.00x	1.45x	1.30x	1.30x	[X]
Monza (Italy)	Temple of Speed. Ultra-low downforce, Engine dominates.	1.75x	0.60x	1.00x	1.00x	[X]
Baku (Azerbaijan)	Monster 2.2km straight meets tight Castle Section.	1.50x	1.00x	1.00x	1.80x	[X]
Singapore (Marina Bay)	Humid night race. Extreme thermal heat & high SC rate.	1.00x	1.00x	1.50x	1.80x	[X]
Austin (COTA, USA)	Turn 1 elevation climb & sweeping S-curves.	1.25x	1.35x	1.20x	1.00x	[OK]
Mexico (Mexico)	High altitude thin air (2,200m). Reduced cooling.	1.35x	0.80x	1.00x	1.00x	[X]
Interlagos (Brazil)	Elevation change, overtaking & rain volatility.	1.30x	1.00x	1.00x	1.00x	[OK]
Las Vegas (USA)	Cold night strip. Long 1.9km straight, low tyre warmup.	1.55x	0.70x	1.20x	1.00x	[X]

Circuit	Description	Engine	Aero	Tyre	SC Mult	Sprint?
Qatar (Lusail)	Ultra high-speed flowing curves & intense G-force heat.	1.00x	1.40x	1.60x	1.00x	[OK]
Abu Dhabi (Yas Marina)	Twilight season finale. Balanced sectors.	1.20x	1.20x	1.10x	1.00x	[X]

[TIP] Strategy Insight: At Monza (Engine 1.75x, Aero 0.60x), investing heavily in Engine upgrades gives a massive advantage while Aero investment is nearly wasted. At Monaco (Aero 1.70x, Engine 0.70x), the opposite is true.

15. AI GRID SYSTEM (20-CAR GRID)

Every Grand Prix features a full **20-car starting grid**. Human players fill slots first; remaining positions are filled by AI drivers modeled after real F1 drivers.

AI Driver Roster (20 Drivers)

#	AI Team	Engine	Aero	Pace	Qual	Consistency
1	Red Bull (Verstappen)	16	16	96	96	94
2	Mercedes (Hamilton)	15	15	94	94	95
3	Ferrari (Leclerc)	16	15	93	97	88
4	McLaren (Norris)	15	16	93	92	91
5	Aston Martin (Alonso)	14	14	91	90	92
6	McLaren (Piastrri)	14	15	90	91	89
7	Mercedes (Russell)	14	14	89	92	87
8	Ferrari (Sainz)	14	14	89	91	86
9	Williams (Albon)	14	11	85	88	84
10	Alpine (Gasly)	12	12	84	83	82
11 - 20	(Midfield & Backmarkers)	8 - 12	8 - 12	70 - 83	71 - 84	71 - 82

AI Rival System

Each human player is assigned an **AI Rival** from the grid. Beating your rival in a GP earns a **+500¢ bonus!**

[NOTE] The AI Rival System and its +500¢ bonus are automatically deactivated if 20 or more human players register for a Grand Prix or Sprint (since all 20 grid slots will be occupied by human players and no AI drivers will be on the grid).

16. WDC SEASON & CHAMPIONSHIP MANAGEMENT

Season Lifecycle

[FLOW DIAGRAM: WDC Season and Championship Lifecycle (Details in guide command registry)]

Season Commands

Command	Description
/season create <name>	Create a new WDC Season

Command	Description
/season add_race <track> [laps]	Add a GP round to the calendar
/season add_sprint_race <track> [laps]	Add a Sprint round to the calendar
/season calendar	View/manage calendar with interactive UI (reorder, remove, start)
/season end	Conclude season and crown World Driver Champion
/season cancel	Cancel the active season

Championship Standings

- Points accumulate across all finished races in the season
- /standings shows current live championship points table
- Champion is determined by total points when season ends

17. ADMIN SUITE & SERVER MANAGEMENT

4-Tier Admin Permission System

Tier	Access Level	Auto-Detected?
Server Owner	Full access to all commands	[OK] Automatic
Discord Administrator	Full access to all commands	[OK] Automatic
Admin Role	Admin commands access	[OK] If user has "Admin" role
Game Admin	Admin commands access	Via /admin addadmin

Admin Command Reference

Command	Description
/admin gp [laps]	Open Grand Prix Admin Panel -- create, manage, run GP events
/admin sprint [laps]	Open Sprint Race Admin Panel -- create, manage, run Sprint events
/admin setstat <user> <stat> <value>	Set any driver skill (1 - 100) or garage part level (1 - 20)
/admin give <user> <amount>	Award credits to a player
/admin remove <user> <amount>	Deduct credits from a player
/admin resetprofile <user>	Completely reset a user's profile to starting defaults
/admin broadcast <message>	Post an announcement embed to any channel (supports @everyone/@here pings)
/admin dbbackup	Create a timestamped backup copy of the SQLite database
/admin addadmin <user>	Grant Game Admin access to a user
/admin removeadmin <user>	Revoke a user's Game Admin access
/admin listadmins	View all users with Game Admin access

GP Admin Panel Features

The /admin gp command opens an interactive control panel with: - **Track Selection Dropdown** -- All 24 official F1 tracks with autocomplete - **Registration Management** -- View registered drivers - **Qualifying Launch** -- Start Q1 -> Q2 -> Q3 knockout qualifying - **Race Launch** -- Begin lap-by-lap race simulation with live telemetry - **Cancel/Reset** -- Cancel active GP with automatic entry fee refund

18. DEEP MATHEMATICAL MODEL & PHYSICS EQUATIONS

Complete Physics Reference Table

Physics Subsystem	Mathematical Formula	Engine Effect
Base Car Power	$P = (\text{Effective_Stat} \times \text{Weight} \times \text{Track_Mod})$	Weighted sum of all 6 parts x track modifiers x rarity bonuses
Lap Time Delta	$T = T_{\text{base}} - (\text{Eng} \times 0.05 + \text{Aero} \times 0.04 + \text{ERS} \times 0.04) - (\text{Pace} / 100 \times 2.0)$	Engine shaves 0.05s/lvl, Aero & ERS shave 0.04s/lvl
Tyre Wear Curve	$\text{Wear} = \text{Base} \times \text{Strategy_Mult} \times \text{Aggression_Mult}$	Base: Soft=12%, Med=7%, Hard=4%. Push=1.4x, Save=0.7x
Wear Penalty	$\text{Penalty} = ((80 - \text{Health})^{1.5}) \times 0.005$	Non-linear. Minimal below 80%, severe below 40%
Damage Power Loss	$\text{Mult} = 1.0 - \min(0.5, \text{Total_Damage} / 1000)$	Gradual power loss, capped at -50%
Overtake Success	$P = 0.45 + (\text{Overtaking-Consistency}) / 200 + \text{Aggro} / 250 + \text{TyreAdv} + \text{PowerAdv}$	Clamped 10% - 95%
Pit Stop Time	$T = 8.0s + (3.5 - \text{Pit_Crew} \times 0.15)$	Level 20 pit crew: 0.5s crew time
Pit Under SC	$T = 3.0s + (3.5 - \text{Pit_Crew} \times 0.15)$	5.0s cheaper under Safety Car
DNF Probability	$\text{Base} = \max(0.2, 6.5 - \text{Reliability} \times 0.25) / \text{Total_Laps}$	Per lap. Modified by aggression, strategy, weather
Spin/Mistake	$P = 0.02 \times (1.5 - \text{Consistency} / 100)$ per lap	Higher consistency = fewer mistakes
Qualifying Speed	$T = \text{Base} - (\text{Car_Power} + \text{Qual} / 2 + \text{Tyre Bonus}) \times 0.12$	Soft tyres give +6.0 performance in quali

19. COMPLETE COMMAND REGISTRY

Player Commands (21)

#	Command	Category	Description
1	/start	[GAME] Onboarding	Create your racing team profile
2	/profile	[CLIPBOARD] Profile	View team profile card (PNG image)
3	/team	[CLIPBOARD] Profile	View detailed team summary
4	/garage	[WRENCH] Garage	View car component levels & damage
5	/upgradeshop	[WRENCH] Garage	Browse upgrade levels & costs
6	/upgrade	[WRENCH] Garage	Upgrade a car part to boost power
7	/inventory	[WRENCH] Garage	View stored parts & equip setups
8	/repairs	[WRENCH] Garage	View damage levels & repair costs
9	/repair	[WRENCH] Garage	Pay credits to repair damage
10	/train	[TRAIN] Training	Train a driver skill (+1 per session)
11	/practice	[TRAIN] Training	Run solo track practice session
12	/daily	[MONEY] Economy	Claim daily login bonus (500¢)
13	/work	[MONEY] Economy	Perform team job for credits
14	/crate	[BOX] Crates	View available loot crates & drop rates
15	/open	[BOX] Crates	Unbox a Rookie, Pro, or Champion crate
16	/shop	[SHOP] Shop	View consumable boosters shop
17	/booster	[SHOP] Shop	Purchase a consumable booster
18	/race	[FLAG] Racing	Challenge another user to a 1v1 duel
19	/bet	[FLAG] Racing	Wager-based race challenge
20	/leaderboard	[CHART] Stats	View server-wide rankings
21	/help	[INFO] Info	Show detailed game guide

Race Event Commands (7)

#	Command	Description
1	/joinrace	Register for upcoming GP (1,000¢ entry fee)
2	/leaverace	Leave GP and receive entry fee refund
3	/grid	View registration list & qualifying grid
4	/standings	View championship points standings
5	/results	View results of last completed GP
6	/strategy	Configure race strategy & tyre compound
7	/pit	Schedule pit stop for next lap

Admin Commands (11)

#	Command	Description
1	/admin gp	Grand Prix Admin Panel
2	/admin sprint	Sprint Race Admin Panel
3	/admin setstat	Set driver/garage stats
4	/admin give	Award credits to player
5	/admin remove	Deduct credits from player
6	/admin resetprofile	Reset player profile to defaults
7	/admin broadcast	Post announcement embed
8	/admin dbbackup	Backup SQLite database
9	/admin addadmin	Grant Game Admin access
10	/admin removeadmin	Revoke Game Admin access
11	/admin listadmins	View all Game Admins

Season Commands (6)

#	Command	Description
1	/season create	Create new WDC Season
2	/season add_race	Add GP round to calendar
3	/season add_sprint_race	Add Sprint round to calendar
4	/season calendar	View & manage season calendar
5	/season end	Conclude season & crown champion
6	/season cancel	Cancel active season

Total: 45 commands (21 Player + 7 Race Event + 11 Admin + 6 Season)

20. DATABASE SCHEMA & ARCHITECTURE

Modular Cogs Architecture

The bot uses Discord.py's **Cogs** extension system for clean modular organization:

Cog	File	Responsibility
AdminCog	cogs/admin.py	Admin controls, season management, GP panels
GarageCog	cogs/garage.py	Profile, team, garage, upgrades, inventory, repairs
EconomyCog	cogs/economy.py	Daily/work, crates, shop, boosters, training, leaderboard, bets
RacingCog	cogs/racing.py	Duels, GP lifecycle, qualifying, live telemetry views

Cog	File	Responsibility
SimulatorCog	cogs/simulator.py	Practice sessions, help command

Database Tables

Table	Purpose
users	Player profiles, credits, XP, level, strategy preferences
drivers	6 driver skill attributes per user
strategists	6 strategist skill attributes per user
garage	6 car component levels + damage tracking
races	Race events (GP/Sprint) with weather, status, track
race_entries	Per-driver race registrations, qualifying times, results
bets	Spectator race betting records
seasons	WDC Season records
season_calendar	Per-season race calendar with ordering
user_inventory	Crate-dropped parts storage with rarity & equipped status
user_boosters	Consumable booster inventory with charge counts
track_mastery	Per-track practice history and pace bonuses
bot_admins	Game admin access grants per server

Technology Stack

Component	Technology
Runtime	Python 3.11+
Framework	discord.py 2.x (app_commands / slash commands)
Database	SQLite 3 (WAL mode, B-Tree indexes)
Rendering	Pillow (PIL) for profile cards, Matplotlib for telemetry charts
Deployment	Docker-ready (Dockerfile included)

DISCORD GRAND PRIX -- *The most advanced Formula 1 simulation engine for Discord.*

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